ASSIGNMENT 2

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BRANCH : MATHEMATICS AND COMPUTING

1) ls (list of files)

Write a C++ program that lists all files and directories in the current working directory.

Hint: Use opendir(), readdir(), closedir().

CODE:

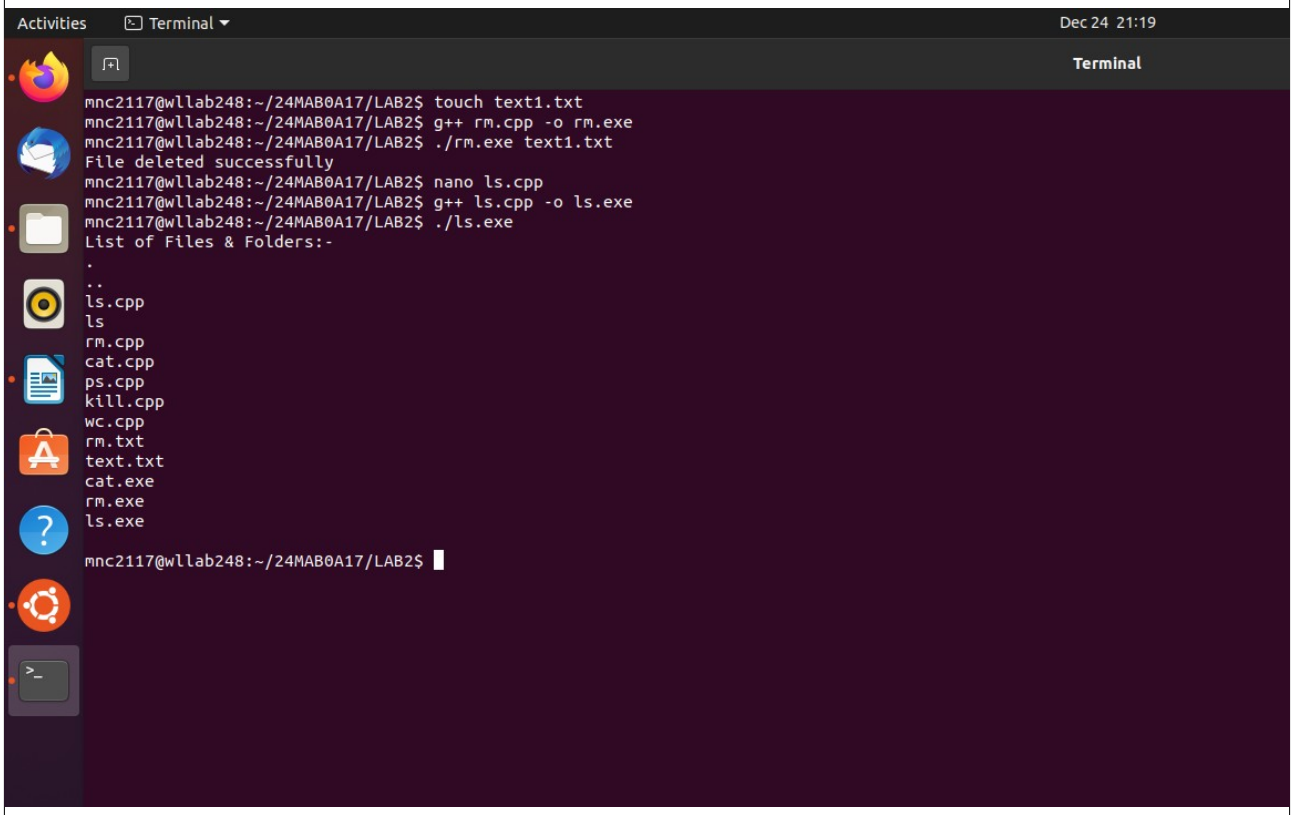
```
#include <iostream>
#include <dirent.h>
using namespace std;

int main() {
	DIR *dir = opendir(".");
	struct dirent *entry;

	while ((entry = readdir(dir)) != NULL)
		cout << entry->d_name << endl;

	closedir(dir);
	return 0;
}
```

OUTPUT:



2) rm(remove files):

Write a C++ program that deletes a file specified by the user.

Hint: Use remove() or unlink().

CODE:

```
#include <iostream>
#include <unistd.h>
using namespace std;

int main() {
  string filename;
  cout << "Enter filename: ";
  cin >> filename;

  if (unlink(filename.c_str()) == 0)
     cout << "File deleted successfully\n";
  else
     cout << "Error deleting file\n";

  return 0;
}
```

OUTPUT:

3) cat(file content):

Write a C++ program that displays the contents of a text file.

Hint: Use fopen(), fgetc(), fread().

CODE:

```
#include <iostream>
#include <cstdio>
using namespace std;

int main() {
char filename[50];
cout << "Enter filename: ";
cin >> filename;

FILE *fp = fopen(filename, "r");
char ch;

while ((ch = fgetc(fp)) != EOF)
  cout << ch;

  fclose(fp);
    return 0;
}
```

OUTPUT:

```
mnc2117@wllab248:~/24MAB0A17/LAB3$ ./ls.exe
.
..
ls.cpp
hi.txt
hii.txt
ls.exe
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano ls.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ touch rm.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano rm.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ g++ rm.cpp -o rm.exe
mnc2117@wllab248:~/24MAB0A17/LAB3$ ./rm.exe hi.txt
Enter filename: hi.txt
File deleted successfully
mnc2117@wllab248:~/24MAB0A17/LAB3$ touch cat.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano cat.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano hii.txt
mnc2117@wllab248:~/24MAB0A17/LAB3$ g++ cat.cpp -o cat.exe
mnc2117@wllab248:~/24MAB0A17/LAB3$ ./cat.exe hii.txt
Enter filename: hii.txt
hi i am saikiran , from mathematics and computing second year . I am 19 years old
mnc2117@wllab248:~/24MAB0A17/LAB3$
```

4) ps(process status):

Write a C++ program that displays currently running process IDs.

Hint: Read information from the /proc directory.

CODE:

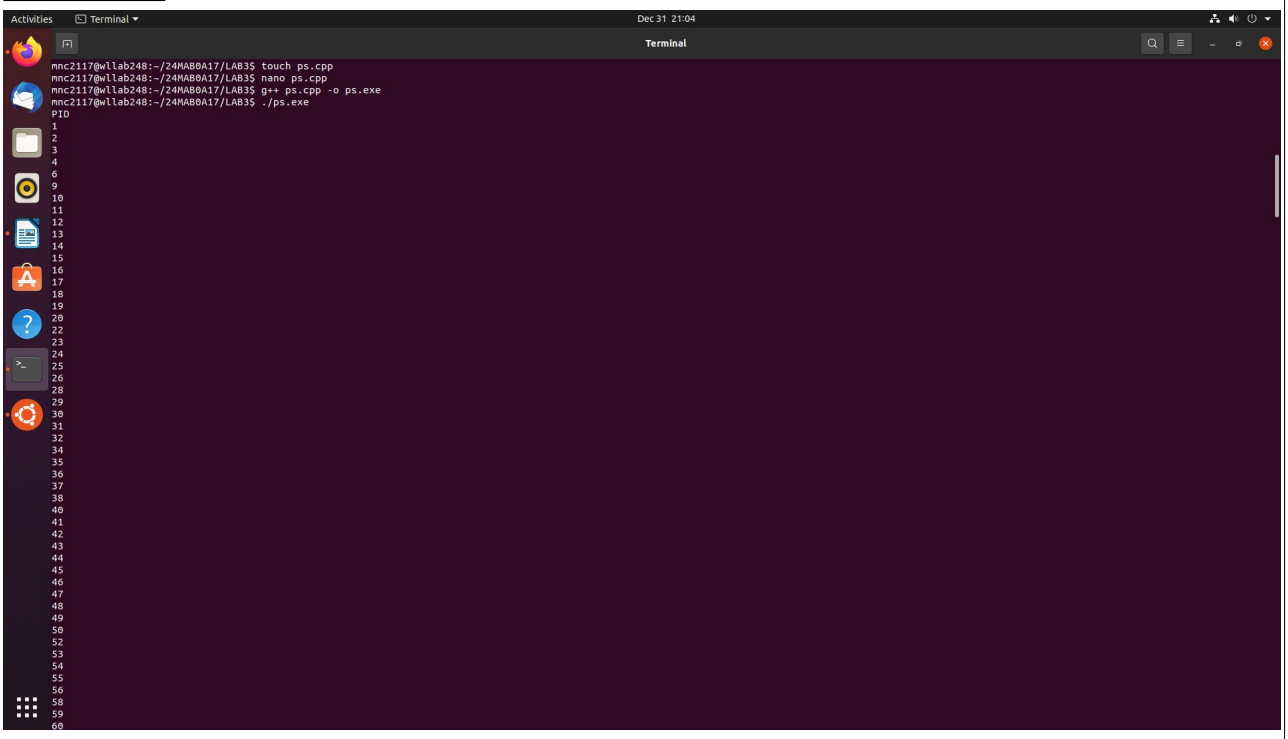
```
#include <iostream>
#include <dirent.h>
#include <cctype>
using namespace std;

int main() {
  DIR *dir = opendir("/proc");
  struct dirent *entry;

  cout << "PID\n";
  while ((entry = readdir(dir)) != NULL)
    if (isdigit(entry->d_name[0]))
  cout << entry->d_name << endl;

    closedir(dir);
    return 0;
}
```

OUTPUT:



5) kill:

Write a C++ program that terminates a process using its PID.

Hint: Use kill() system call with SIGKILL or SIGTERM.

CODE:

```
#include <iostream>
#include <signal.h>
using namespace std;

int main() {
  int pid;
  cout << "Enter PID: ";
  cin >> pid;

  kill(pid, SIGKILL);
    cout << "Process terminated successfully\n";
  return 0;
}
```

OUTPUT:

6) wc :

Write a C++ program that counts lines, words, and characters in a file.

CODE:

```
#include <iostream>
#include <fstream>
#include <sstream>
using namespace std;

int main() {
  string filename, line, word;
  int lines = 0, words = 0, chars = 0;
cout << "Enter filename: ";
    cin >> filename;
ifstream file(filename);
  while (getline(file, line)) {
        lines++;
    chars += line.length();
        stringstream ss(line);
    while (ss >> word) words++;
  }

  cout << "Lines: " << lines << endl;
    cout << "Words: " << words << endl;
    cout << "Characters: " << chars << endl;
    return 0;
}
```

OUTPUT:

```
10246
10322
10361
10366
10453
mnc2117@wllab248:~/24MAB0A17/LAB3$ touch kill.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano kill.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ g++ kill.cpp -o kill.exe
mnc2117@wllab248:~/24MAB0A17/LAB3$ ./kill.exe
Enter PID: 23
Process terminated successfully
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano kill.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano kill.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ touch wc.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ nano wc.cpp
mnc2117@wllab248:~/24MAB0A17/LAB3$ g++ wc.cpp -o wc.exe
mnc2117@wllab248:~/24MAB0A17/LAB3$ ./wc.exe
Enter filename: hii.txt
Lines: 1
Words: 17
Characters: 82
mnc2117@wllab248:~/24MAB0A17/LAB3$
```